

Linux Privilege Escalation Automation Toolkit

Detailed Project Report

Abstract

This project focuses on building an automated Linux Privilege Escalation Automation Toolkit that identifies common privilege escalation weaknesses through safe enumeration and reporting. The toolkit performs detection only and does not exploit vulnerabilities. This project focuses on building an automated Linux Privilege Escalation Automation Toolkit that identifies common privilege escalation weaknesses through safe enumeration and reporting. The toolkit performs detection only and does not exploit vulnerabilities. This project focuses on building an automated Linux Privilege Escalation Automation Toolkit that identifies common privilege escalation weaknesses through safe enumeration and reporting. The toolkit performs detection only and does not exploit vulnerabilities. This project focuses on building an automated Linux Privilege Escalation Automation Toolkit that identifies common privilege escalation weaknesses through safe enumeration and reporting. The toolkit performs detection only and does not exploit vulnerabilities.

Introduction

Privilege escalation is one of the most important stages in penetration testing and post-exploitation. Linux systems frequently contain misconfigurations such as SUID binaries, weak permissions, cron vulnerabilities, insecure services, and outdated kernels. Privilege escalation is one of the most important stages in penetration testing and post-exploitation. Linux systems frequently contain misconfigurations such as SUID binaries, weak permissions, cron vulnerabilities, insecure services, and outdated kernels. Privilege escalation is one of the most important stages in penetration testing and post-exploitation. Linux systems frequently contain misconfigurations such as SUID binaries, weak permissions, cron vulnerabilities, insecure services, and outdated kernels. Privilege escalation is one of the most important stages in penetration testing and post-exploitation. Linux systems frequently contain misconfigurations such as SUID binaries, weak permissions, cron vulnerabilities, insecure services, and outdated kernels.

Objectives

Develop an automated scanner, identify privilege escalation vectors, generate structured reports, and provide mitigation guidance. Develop an automated scanner, identify privilege escalation vectors, generate structured reports, and provide mitigation guidance. Develop an automated scanner, identify privilege escalation vectors, generate structured reports, and provide mitigation guidance. Develop an automated scanner, identify privilege escalation vectors, generate structured reports, and provide mitigation guidance.

System Architecture

The toolkit follows a modular architecture consisting of System Information Collection, SUID Scanner, Permission Scanner, Service Scanner, Cron Scanner, Kernel CVE Scanner, Analysis Engine, and Report Generator. The toolkit follows a modular architecture consisting of System Information Collection, SUID Scanner, Permission Scanner, Service Scanner, Cron Scanner, Kernel CVE Scanner, Analysis Engine, and Report Generator. The toolkit follows a modular architecture consisting of System Information Collection, SUID Scanner, Permission Scanner, Service Scanner, Cron Scanner, Kernel CVE Scanner, Analysis Engine, and Report Generator. The toolkit follows a modular architecture consisting of System Information Collection, SUID Scanner, Permission Scanner, Service Scanner, Cron Scanner, Kernel CVE Scanner, Analysis Engine, and Report Generator.

Module 1 – System Information

Collects hostname, OS version, kernel version, user context, PATH variables, environment variables, and network information. Collects hostname, OS version, kernel version, user context, PATH variables, environment variables, and network information. Collects hostname, OS version, kernel version, user context, PATH variables, environment variables, and network information. Collects hostname, OS version, kernel version, user context, PATH variables, environment variables, and network information.

Module 2 – SUID/SGID Scanner

Discovers SUID and SGID binaries and compares them against known GTFOBins techniques to identify high-risk binaries. Discovers SUID and SGID binaries and compares them against known GTFOBins techniques to identify high-risk binaries. Discovers SUID and SGID binaries and compares them against known GTFOBins techniques to identify high-risk binaries. Discovers SUID and SGID binaries and compares them against known GTFOBins techniques to identify high-risk binaries.

Module 3 – Permission Scanner

Detects world-writable files, directories, exposed home folders, and sensitive file permission issues. Detects world-writable files, directories, exposed home folders, and sensitive file permission issues. Detects world-writable files, directories, exposed home folders, and sensitive file permission issues. Detects world-writable files, directories, exposed home folders, and sensitive file permission issues.

Module 4 – Service Scanner

Reviews systemd services, dangerous sudo rules, writable service binaries, and suspicious root-owned processes. Reviews systemd services, dangerous sudo rules, writable service binaries, and suspicious root-owned processes. Reviews systemd services, dangerous sudo rules, writable service binaries, and suspicious root-owned processes. Reviews systemd services, dangerous sudo rules, writable service binaries, and suspicious root-owned processes.

Module 5 – Cron Scanner

Enumerates cron jobs and detects writable scripts, wildcard injection opportunities, and PATH hijacking issues. Enumerates cron jobs and detects writable scripts, wildcard injection opportunities, and PATH hijacking issues. Enumerates cron jobs and detects writable scripts, wildcard injection opportunities, and PATH hijacking issues. Enumerates cron jobs and detects writable scripts, wildcard injection opportunities, and PATH hijacking issues.

Module 6 – Kernel CVE Scanner

Collects kernel information and matches versions against a local CVE database to identify known privilege escalation risks. Collects kernel information and matches versions against a local CVE database to identify known privilege escalation risks. Collects kernel information and matches versions against a local CVE database to identify known privilege escalation risks. Collects kernel information and matches versions against a local CVE database to identify known privilege escalation risks.

Report Generation

Produces TXT and JSON reports summarizing findings, severity levels, affected resources, and mitigation recommendations. Produces TXT and JSON reports summarizing findings, severity levels, affected resources, and mitigation recommendations. Produces TXT and JSON reports summarizing findings, severity levels, affected resources, and mitigation recommendations. Produces TXT and JSON reports summarizing findings, severity levels, affected resources, and mitigation recommendations.

Technologies Used

Python, Bash, Linux Commands, Git, GitHub, Systemd, Cron, GTFOBins, CVE References. Python, Bash, Linux Commands, Git, GitHub, Systemd, Cron, GTFOBins, CVE References. Python, Bash, Linux Commands, Git, GitHub, Systemd, Cron, GTFOBins, CVE References. Python, Bash, Linux Commands, Git, GitHub, Systemd, Cron, GTFOBins, CVE References.

Testing Methodology

The toolkit was tested by executing scans, validating module outputs, generating reports, and reviewing findings. The toolkit was tested by executing scans, validating module outputs, generating reports, and reviewing findings. The toolkit was tested by executing scans, validating module outputs, generating reports, and reviewing findings. The toolkit was tested by executing scans, validating module outputs, generating reports, and reviewing findings.

Security Considerations

The toolkit is detection-only and is intended for authorized security assessments, educational purposes, and defensive auditing. The toolkit is detection-only and is intended for authorized security assessments, educational purposes, and defensive auditing. The toolkit is detection-only and is intended for authorized security assessments, educational purposes, and defensive auditing. The toolkit is detection-only and is intended for authorized security assessments, educational purposes, and defensive auditing.

Results

The toolkit successfully generated reports and enumerated Linux privilege escalation vectors using modular scanning techniques. The toolkit successfully generated reports and enumerated Linux privilege escalation vectors using modular scanning techniques. The toolkit successfully generated reports and enumerated Linux privilege escalation vectors using modular scanning techniques. The toolkit successfully generated reports and enumerated Linux privilege escalation vectors using modular scanning techniques.

Future Enhancements

HTML reports, CVSS scoring, live CVE feeds, dashboard interface, Docker deployment, and GitHub Actions integration. HTML reports, CVSS scoring, live CVE feeds, dashboard interface, Docker deployment, and GitHub Actions integration. HTML reports, CVSS scoring, live CVE feeds, dashboard interface, Docker deployment, and GitHub Actions integration. HTML reports, CVSS scoring, live CVE feeds, dashboard interface, Docker deployment, and GitHub Actions integration.

Conclusion

The project demonstrates practical Linux security auditing, privilege escalation enumeration, and cybersecurity reporting skills. The project demonstrates practical Linux security auditing, privilege escalation enumeration, and cybersecurity reporting skills. The project demonstrates practical Linux security auditing, privilege escalation enumeration, and cybersecurity reporting skills. The project demonstrates practical Linux security auditing, privilege escalation enumeration, and cybersecurity reporting skills.

References

GTFOBins, MITRE CVE Database, NVD, Linux Kernel CVEs, HackTricks, PEASS-ng Documentation.